HVDC INTERCONNECTS

Design Basis, Design Basis Accident, and Military Action Extension

DBA-ES-TX-FC3: HVDC Interconnect Facility Class

A Child Document Under

DBA-ES: Electric Sector Design Basis Framework (v0.9)

DBA-EN-SF: Energy Sector Framework (v0.8)

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Section 1 — Purpose, Scope, and Inheritance

1.1 Purpose

This document establishes the Design Basis for high-voltage direct current (HVDC) interconnects within the Electric Sector Design Basis Series. It is the third facility-class instrument in the series, following DBA-ES-HY-FC1 (large hydroelectric storage facilities) and DBA-ES-TX-FC2 (large EHV transmission substations).

The document applies the DB → DBA → DBA-MA three-layer analytical progression to HVDC interconnects serving a cross-interconnection transfer role: a baseline Design Basis (DB) establishing postulated events from equipment failure, natural hazards, and operational conditions; a Design Basis Accident (DBA) layer bounding the non-adversarial consequence envelope; and a military action extension (DBA-MA) applying the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to each postulated event category.

HVDC interconnects occupy a unique position in the electric sector's consequence structure. Unlike all other facility classes in the DBA-ES series, whose consequences propagate within a single interconnection, HVDC interconnects carry consequences across interconnection boundaries. The loss of a major HVDC link removes the transfer capability that allows one interconnection to support another under stress — a consequence with no analog in AC transmission and no precedent in the non-adversarial design basis frameworks that govern electric sector planning.

This document introduces Transfer Capability Isolation (TCI) as the governing consequence-duration metric for FC3. TCI is defined as the duration during which cross-interconnection transfer capability is unavailable following loss or block of the HVDC link, expressed in hours, days, or months. TCI complements RT_HVLLC: RT_HVLLC governs the equipment replacement timeline following physical damage to converter transformers; TCI governs the transfer capability consequence regardless of whether physical equipment damage is involved.

1.2 Facility Class Definition

Technology class. The facility is a high-voltage direct current transmission system using line-commutated converter (LCC) or voltage-source converter (VSC) technology.

Transfer capacity. The HVDC link's nameplate transfer capacity equals or exceeds 2,000 MW in either transfer direction.

Cross-interconnection role. The HVDC link connects two separate AC interconnections or sub-regions whose AC systems are not synchronously coupled. In North America, this includes links between the Eastern Interconnection, Western Interconnection, and ERCOT, as well as links to Canadian provincial systems operating asynchronously.

1.3 The TCI Metric

Transfer Capability Isolation (TCI) is introduced in this instrument as the primary consequence-duration metric for FC3. TCI is defined as the continuous duration, measured from the initiating event, during which the HVDC link is unable to deliver its design transfer capacity in the governing transfer direction, expressed in hours (for operational events), days (for equipment repair events), or months (for replacement-class events).

TCI is a functional metric, not a physical one. It measures the loss of the transfer capability function, not the loss of specific equipment. For FC3, the most severe TCI scenario is one in which TCI equals RT_HVLLC — a physical attack on the converter transformer that makes the link unavailable for the full component replacement period of twelve to eighteen months.

1.4 Evidentiary Anchors

Nelson River HVDC System (Manitoba, Canada). The Nelson River bipolar HVDC system — comprising Bipole I (1,620 MW), Bipole II (2,000 MW), and Bipole III (2,000 MW) — is the primary reference case for cross-interconnection HVDC consequence analysis in North America. The prolonged debate over Bipole III routing — driven by concerns about the consequence of a single corridor fault taking down both Bipole I and Bipole II simultaneously — is a documented case study in HVDC corridor concentration risk.

2003 Northeast Blackout (August 14–15, 2003). The 2003 Northeast Blackout demonstrated that the loss of transfer capability between regions, under conditions of prior constraint, transforms a local contingency into a regional cascade. The interaction pattern — compound consequence of simultaneous physical and communications degradation — is the primary analytical framework for FC3's CE consequence analysis.

1.5 Inherited Methodology

All sector-level methodology is inherited from DBA-ES: Electric Sector Design Basis Framework (v0.9). The following table identifies inherited elements and FC3-specific adaptations. The governing foundational instruments — FD-BL (placement), FD-BL-D1 (mode of discharge), FD-BL-D2 (indication integrity), and FD-EV (envelope) — are applied in this annex by reference. Below-line ML "operating envelopes" here are the determinism carve-out region (FD-BL §4.4 / FD-LD), retained as written; FD-EV reserves "envelope" for the pre-authorized sense and renames the monitored operating-limit sense to "operating band."

Element	Source	FC3-Specific Adaptation
DB → DBA → DBA-MA three-layer progression	DBA-ES §3.1	Inherited without modification
Postulated event categories: PE, SE, CE, XC	DBA-ES §3.5	FC3-specific events instantiated in Section 3

Bounding event methodology	DBA-ES §4.1	Converter transformer is governing long-lead-time equipment; TCI is the primary consequence-duration metric.
Natural hazard compound-event requirement	DBA-ES §4.3	Geomagnetic disturbance + seismic as governing compound case
Three-tier consequence taxonomy	DBA-ES §6.1	Cascade pathway governs; TCI defines Tier 2 duration; Tier 3 through regional isolation.
RT_HVLLC extended consequence regime	DBA-ES §6.2	RT_HVLLC secondary to TCI; converter transformer lead times 12–18 months
MA extension trigger criteria	DBA-ES §1.5	FC3 satisfies Triggers 1, 2, 3, and 5 by definition
Four MA characteristics	DBA-ES §5.2	Applied to each FC3 postulated event category in Section 5
Dual Design Basis architecture	DBA-ES §3.4	Physical DB (this document) + AI Governance DB interaction points in §3.6
MA applicability four-filter framework	DBA-ES-MA-SF v0.2 §1.4; DBA-ES-MA-QB v0.1 §5	Instantiated in §5.1 of this document
MA communications design basis (CB-1 through CB-5)	DBA-ES-MA-QB v0.1 §5	Instantiated in §5.5 of this document; inter-terminal communications dependency noted

1.6 Position in Hierarchy

Document Number: DBA-ES-TX-FC3-001

Series: Electric Sector Design Basis Series — Facility Class Instruments

Parent Document: DBA-ES: Electric Sector Design Basis Framework (v0.9)

Series Grandparent: DBA-EN-SF: Energy Sector Framework (v0.8)

Position in hierarchy: Facility-class instrument. All sector-level methodology is inherited from DBA-ES. This document applies that methodology to HVDC interconnects and introduces the TCI consequence-duration metric specific to this facility class.

Section 2 — Facility Profile: HVDC Interconnect

2.1 Physical Configuration

A point-to-point HVDC interconnect consists of two converter stations — a rectifier (AC-to-DC) and an inverter (DC-to-AC) — connected by a DC transmission line or cable. Each converter station contains converter transformers, thyristor or IGBT valve assemblies, smoothing reactors, DC filters, AC filters, and the control and protection systems that manage the conversion process.

The converter transformer is the critical element that defines the FC3 consequence profile. Converter transformers are custom-designed for HVDC service: they must withstand DC voltage stress on the valve-side windings, harmonic currents, and the electrical stresses unique to the commutation process. These requirements produce a design that cannot be substituted with a standard power transformer. The custom design requirement, combined with limited global manufacturing capacity, produces the RT_HVLLC characteristic that makes physical damage to the converter transformer the governing consequence event.

2.2 Grid Topology Role

An HVDC interconnect in a cross-interconnection role performs a function that has no AC equivalent: it allows power to flow between two AC systems that are not synchronously coupled. This asynchronous transfer function provides three grid services simultaneously: energy exchange between interconnections with complementary supply and demand profiles; a transfer capability resource that either interconnection can call upon under stress; and electrical isolation between the two interconnections, preventing fault propagation across the DC link.

The isolation function is both an asset and a vulnerability. As an asset, faults in one interconnection cannot directly propagate to the other. As a vulnerability, the loss of the HVDC link removes transfer capability with no electrical recovery path. The HVDC link is the only transfer mechanism, and its loss is absolute in a way that AC line loss is not.

2.3 Regulatory Context

HVDC interconnects are subject to NERC reliability standards as Transmission Owners and Transmission Operators. However, HVDC systems present regulatory classification challenges because their operating characteristics do not map cleanly onto the AC transmission standards that NERC developed for synchronous generators and AC lines. No mandatory physical security standard equivalent to NERC CIP-014 has been established specifically for HVDC converter stations. This document supplies the design basis analysis that a physical security standard for HVDC would need to draw upon.

Section 3 — Design Basis: FC3 Postulated Event Categories

3.1 Sector-Level Inheritance

The four sector-level postulated event categories — PE, SE, CE, and XC — are inherited from DBA-ES §3.5 and instantiated below with FC3-specific events.

3.2 Category PE — Power Supply and Grid Interface Events

Code	Event	Initiating Condition	Governing Concern
PE-1	AC fault causing forced converter block.	A fault in the AC system at either terminal causes the HVDC control system to block converters. Power transfer ceases instantaneously.	Instantaneous loss of the full HVDC transfer block initiates simultaneous frequency and voltage transients in both sending and receiving AC systems. TCI begins at the block and ends at the restart.
PE-2	Commutation failure at the inverter terminal	Inverter thyristors fail to transfer current correctly, typically due to AC voltage depression at the inverter terminal.	Repeated commutation failures can force a sustained block without physical equipment damage. Recovery depends on AC voltage restoration.
PE-3	DC line fault	Fault on DC transmission line from lightning, insulator flashover, conductor contact, or physical damage.	A bipolar fault reduces transfer capability to zero; a monopolar fault reduces it to approximately half. DC line repair time governs TCI for physical damage.
PE-4	Converter station auxiliary power loss	Loss of AC auxiliary power at either station removes supply for cooling systems, valve hall ventilation, and control electronics.	Converters must be blocked to prevent thermal damage. FC3 analog of PE-SS in FC2. Recovery requires auxiliary power restoration before converter restart.
PE-5	AC network reconfiguration — inadmissible operating point	An AC network topology change moves one or both converter terminals outside the HVDC control system's stable operating range.	System interaction event unique to HVDC: a change at the remote terminal can force a block at the local terminal without any local physical event.

3.3 Category SE — Structural and Physical Integrity Events

Code	Event	Initiating Condition	Governing Concern
SE-1	Converter transformer failure — internal fault	Internal fault within the converter transformer winding or bushing, resulting in differential relay operation.	RT_HVLLC governs. Custom-designed converter transformers have lead times of 12–18 months. Unlike standard EHV transformers, they cannot be sourced from STEP. No strategic reserve exists.
SE-2	Converter transformer failure — external	Physical damage from fire, flooding, projectiles, or blast. Oil-filled tank and external bushings are the primary	Bounding SE event. Combines RT_HVLLC consequence of converter transformer replacement with TCI consequence of link unavailability. Compound metric: TCI = RT_HVLLC = 12–18

	physical damage	vulnerable elements.	months.
SE-3	Thyristor valve damage or failure	Physical or electrical damage to thyristor valve assemblies in the converter hall.	Thyristor valve replacement is faster than a converter transformer (4–8 weeks). Physical damage to the valve hall structure, compromising environmental controls, extends the repair timeline.
SE-4	Geomagnetic disturbance (GMD)	Severe GMD induces geomagnetically induced currents (GIC) in long DC transmission lines and converter transformers.	HVDC systems are particularly susceptible to GIC relative to AC transmission. A Carrington-class event is the governing natural hazard for FC3. GMD at both terminals simultaneously produces TCI = RT_HVLLC at each terminal.
SE-5	Compound natural hazard	Sequential natural hazard events. Governing case: GMD affecting converter transformer condition, followed by a seismic event at one terminal.	GMD-seismic sequence exploits reduced dielectric and thermal margins of the GMD-stressed transformer. Establishes the upper bound of the non-adversarial extended consequence regime.

3.4 Category CE — Control and Communications Events

Control and communications events at FC3 have a structural characteristic unique in the DBA-ES series: HVDC control systems require continuous inter-terminal communication as a functional requirement of normal operation. CE events that sever or degrade inter-terminal communications may directly force a converter block or restrict the link to a degraded operating mode.

Code	Event	Initiating Condition	Governing Concern
CE-1	Loss of HVDC inter-terminal control communications	Loss of telecommunications link between rectifier and inverter converter stations.	Unlike AC transmission, HVDC requires active inter-terminal communication for normal operation. Communications failure may force link block or limit link to fixed-power degraded mode. Inter-terminal comms is a critical single-point dependency with no AC analog.
CE-2	HVDC control system malfunction or cyber intrusion	Malfunction of or unauthorized access to the HVDC control system, including power order manipulation or firing angle modification.	HVDC control systems are a high-value cyber target: manipulation can produce large, instantaneous power flow changes in both AC networks without physical equipment damage.
CE-3	Loss of CntlCo SCADA visibility for DC power flow	Loss of SCADA visibility into HVDC link status, power flow, and converter station condition at the reliability coordinator or BA level.	CntlCo cannot correctly account for DC transfer capability in AC system management decisions. Incorrect contingency analysis can lead to operating the AC system in an unstable configuration following an actual HVDC block.

3.5 Category XC — Cross-Sector Cascade Events

Cross-sector cascade events at FC3 operate at a geographic scale exceeding all other facility classes in the DBA-ES series. While FC2 cascade events affect a metropolitan load pocket, FC3 cascade events affect an entire interconnection or large sub-region. XC-4 is identified at the DB layer — not only in the MA extension — because the timing dependency is a structural characteristic of the HVDC facility class that exists independent of adversarial intent.

Code	Event	Initiating Condition	Governing Concern
XC-1	Regional isolation through cross-interconnection transfer loss	Loss of the HVDC link removes the transfer capability that allows one interconnection to support another under stress.	Defining XC event for FC3. The 2003 Northeast Blackout demonstrated how the loss of transfer capability transforms a local event into a cross-regional cascade. TCI metric bounds this consequence.
XC-2	Frequency instability in the islanded receiving region	Receiving interconnection experiences an instantaneous generation deficit equal to the lost HVDC transfer block.	If the deficit exceeds the available frequency response, UFLS relays operate. For a 3,000 MW block into a low-inertia system, cascading UFLS may initiate before any operator action is possible.
XC-3	Cross-sector consequence through regional power deficit	Regional power deficit following HVDC block or extended TCI produces load shedding, removing power from cross-sector critical facilities.	Tier 3 pathway for FC3. Requires critical load profile inventory in the receiving region and backup posture analysis for most time-constrained critical facilities.
XC-4	Attack timed to peak import dependency.	Event — adversarial or non-adversarial — occurs during maximum HVDC import dependency in the receiving system, maximizing the generation deficit to be absorbed.	Identified at the DB layer because the timing dependency is a structural HVDC vulnerability independent of adversarial intent. The same physical event produces materially worse consequences at peak import than at off-peak.

3.6 Dual Design Basis Architecture

Consistent with DBA-ES §3.4, an FC3 facility simultaneously carries a physical Design Basis and an AI Governance Design Basis. Three AI governance interaction points are established for FC3, closing DI-5.

HVDC control optimization — Bright Line: below, with operating envelope boundary requirement. ML-based power flow optimization and commutation failure prediction tools operate below the Bright Line within their trained operating envelopes. Bright Line condition: operating envelope must include explicit margins to commutation failure voltage thresholds and AC stability limits at both terminals. ML optimization must not autonomously adjust power order or firing angle in ways that reduce these margins without a human-authorized envelope update. Compromised ML through adversarial data poisoning can produce control outputs that destabilize both terminal AC systems simultaneously.

Anomaly detection for cyber intrusion — Bright Line: below, with adversarial input robustness requirement. ML-based anomaly detection operates below the Bright Line. FC3 convergence of MA scenario and AI governance attack surface: an adversary pre-positioned in the HVDC control system may manipulate the sensor feeds the anomaly detection ML monitors, blinding detection before the operational attack. Anomaly detection models must be validated against adversarial input scenarios — not only random sensor noise — and sensor feeds must use paths architecturally independent of the HVDC control command path.

Gen AI operator support for HVDC operations — Bright Line: above; Command Broker required. Gen AI advisory tools for HVDC operations are non-deterministic and operate above the Bright Line. Command Broker is the qualified HVDC system operator or designated CntlCo authority with power order authority. HVDC-specific Command Broker constraint: power order changes affect AC frequency at both terminals simultaneously. The Command Broker must evaluate the AC frequency trajectory at both terminals before authorizing any Gen AI-recommended power order change during a grid emergency.

Section 4 — Design Basis Accidents: Non-Adversarial Envelope

4.1 Bounding Event Methodology

For FC3, two independent criteria determine the bounding event: the event producing the longest TCI (SE-2, converter transformer external physical damage — RT_HVLLC of 12–18 months), and the event producing the most severe instantaneous AC transient (PE-1 at maximum transfer rate). Both bounding events must be analyzed separately.

4.2 Non-Adversarial Consequence Envelope

Tier 1 — Facility-level. Loss of HVDC transfer capability for RT_HVLLC duration (12–18 months) following converter transformer replacement-class damage. Monopolar operation at half capacity may be available if only one pole's converter transformer is damaged.

Tier 2 — Grid-level. Instantaneous frequency and voltage transient in the receiving AC system at the maximum transfer block, followed by a sustained regional power deficit for the TCI duration. The instantaneous transient is bounded by the receiving system's frequency response and UFLS relay settings.

Tier 3 — Cross-sector. Load shedding in the receiving region during peak import dependency, removing power from defense-critical and other critical facilities for a duration determined by TCI and critical facilities' backup power posture.

4.3 Extended Consequence Regime

The extended consequence regime for FC3 is defined by the intersection of TCI and the receiving region's resource adequacy posture during the TCI period. A TCI of 12–18 months covers at least one full peak demand season. Resource adequacy planning must demonstrate that the receiving interconnection can sustain reliability obligations without HVDC import contribution for this duration.

The extended consequence regime analysis must address supply chain interdiction risk. Unlike large AC power transformers — which have at least some domestic manufacturing — HVDC converter transformers are manufactured by a very small number of global suppliers, none of which are domestic US manufacturers for units at the relevant voltage and power class.

Section 5 — Military Action Extension

5.1 MA Extension Applicability — Four-Filter Assessment

Per DBA-ES-MA-SF v0.2 §1.4 and DBA-ES-MA-QB v0.1 §5, each DBA-ES FC instrument must present a structured four-filter applicability assessment before the operator engages with the MA extensions. For FC3, the assessment is resolved at the class level as follows.

- **Filter 1 — Geolocation relative to defense-critical loads:** SATISFIED BY DEFINITION. HVDC interconnects serve the receiving interconnection's entire load, which by definition includes defense-critical loads, military installations, and continuity-of-government facilities. XC-3 (cross-sector consequence through regional power deficit) is an explicitly identified postulated event category for FC3.
- **Filter 2 — Position in strategic transmission corridors:** SATISFIED BY DEFINITION. FC3 facility class criteria require a cross-interconnection transfer role. A cross-interconnection HVDC link is definitely the only transfer corridor between the two interconnections it serves. Loss of an FC3 link is an absolute corridor loss with no AC recovery path.
- **Filter 3 — Substitutability:** NONE. The HVDC link's substitutability is zero at the interconnection boundary level: there is no alternative mechanism for cross-interconnection power transfer if the HVDC link is blocked. Converter transformer RT_HVLLC of 12–18 months with no STEP program equivalent and no domestic US manufacturer at the governing voltage class makes the substitutability gap the most severe in the DBA-ES series.
- **Filter 4 — Adversary targeting logic:** HIGHEST PRIORITY TARGET IN THE SERIES. DBA-ES-MA-SF v0.2 §4.2 identifies FC3 as a member of both high-consequence pattern attack combinations involving transmission infrastructure (FC3+CntlCo; FC2+FC3). Regional isolation through HVDC targeting is a strategic military objective — it removes an interconnection's external support infrastructure without requiring attacks on every generator in the region.

Conclusion: The MA design basis applies to all FC3 facilities. No facility-level exemption assessment is required. The MA extension sections of this instrument apply without a threshold qualification.

5.2 Application of the Four Characteristics

Category	Governing MA Scenario	Four Characteristics Applied	TCI / RT_HVLLC Implication	Acceptance Criterion
PE	Coordinated a cyber-physical attack, forcing a sustained converter block at both terminals simultaneously.	Intentionality: targets the control system at both terminals to prevent automatic restart. Persistence: control system compromise sustains	TCI begins at block initiation. Duration governed by control system recovery under adversarial persistence — potentially days to weeks if access is	Inter-terminal communications must have an independent, physically separated backup path not subject to the same intrusion vector as the primary

		block beyond normal recovery. Precision: targets inter-terminal communications — the single dependency whose loss prevents restart. Denial: Physical access denial prevents manual override.	denied. RT_HVLLC is not engaged unless physical equipment damage accompanies.	path.
SE	Precision kinetic attack on converter transformer at one or both terminals designed to maximize RT_HVLLC	Intentionality: selects converter transformer as the highest-RT_HVLLC element. Persistence: 12–18 months lead time with no STEP equivalent is adversary's persistence without further action. Precision: oil-filled bushings and external radiators targeted. Denial: Access denial prevents damage assessment and replacement delivery.	TCI = RT_HVLLC = 12–18 months. Both metrics are simultaneously engaged. Governing consequence case for FC3.	Physical hardening of the converter transformer is required. Absence of STEP equivalent for converter transformers is a strategic gap requiring national-level policy response (DI-2).
CE	Cyber intrusion inducing forced power reversal or rapid uncontrolled power order change.	Intentionality: studies HVDC control architecture to identify power order manipulation, maximizing AC instability at the receiving terminal. Persistence: control system compromise persists until forensically remediated. Precision: targets a specific control parameter causing instability without an obvious signature. Denial: SCADA loss removes visibility simultaneously.	TCI during an active cyber event is determined by the operator's ability to detect anomalous behavior and manually block it. Post-event TCI determined by control system forensic investigation — potentially weeks.	HVDC control system must include anomaly detection for uncharacteristic power order changes, with automatic protective block authority independent of compromised control pathway.
XC	Attack timed to peak import dependency — coordinated with energy market intelligence.	Intentionality: times attack to the hour of maximum HVDC import dependency in the receiving system. Persistence:	TCI consequence severity is a function of both duration and timing. An 18-month TCI beginning during peak summer import	Regional resource adequacy planning must treat HVDC import as a contingency resource, not a firm resource, demonstrating adequacy

		RT_HVLLC sustains deficit through peak demand season. Precision: Attack timing is the primary precision element. Denial: Access denial prevents emergency generator dispatch coordination.	produces qualitatively more severe consequences than the same TCI beginning in a low-demand period.	without HVDC contribution for RT_HVLLC duration.

5.3 The Regional Isolation Strategy

The governing MA scenario for FC3 is qualitatively different from any scenario in FC2. At FC2, the adversary's objective is to deprive a metropolitan load pocket of power. At FC3, the adversary's objective is to isolate a region from its external support infrastructure — to remove the interconnection's ability to call upon neighboring systems under stress, at a time when the region is most dependent on that capability. Regional isolation through HVDC targeting is a strategic military objective, not a tactical one.

This strategic character is the reason XC-4 is identified at the DB layer. The timing dependency is a structural vulnerability of the HVDC facility class independent of adversarial intent: a converter transformer failure in July during peak summer import produces worse consequences than the same failure in April. The MA extension adds intentionality to the timing — the adversary chooses July — but the underlying vulnerability is present in the non-adversarial design basis.

5.4 Supply Chain as a Persistence Vector

The absence of a strategic reserve for HVDC converter transformers and the concentration of global manufacturing in a small number of non-domestic suppliers creates a supply chain persistence vector with no equivalent in other facility classes. An adversary who attacks an HVDC converter transformer and simultaneously interdicts the supply chain for replacement units extends TCI beyond nominal RT_HVLLC without any further direct action against the facility. Supply chain interdiction as an MA persistence vector is not hypothetical: the concentration of converter transformer manufacturing creates discrete, identifiable choke points in the replacement supply chain.

5.5 MA Communications Design Basis

DBA-ES-MA-QB v0.1 §5 establishes the MA communications design basis as a sector-level requirement applicable to all DBA-ES FC instruments. The MA communications baseline condition — primary SCADA and voice communications channels are unavailable — is the assumed design basis condition for all MA scenario analysis in this instrument. The following instantiates CB-1 through CB-5 for FC3, with a note specific to FC3's unique inter-terminal communications dependency.

FC3-specific communications dependency note: Unlike all other DBA-ES FC instruments, FC3 facilities require inter-terminal communications as a functional prerequisite for HVDC power transfer (CE-1 postulated event). The CB-1 backup channel requirement, therefore, applies in two distinct senses at FC3: (a) backup channel from each converter station to the serving CntlCo (standard CB-1 obligation), and (b) backup inter-terminal channel between the two converter stations themselves (FC3-specific obligation). Loss of both primary and backup inter-terminal channels forces a converter block regardless of physical equipment status. The CB-1 design must address both channel types explicitly.

- **CB-1 (Backup channel):** Each FC3 converter station must maintain at least one backup communications channel to the serving CntlCo, architecturally independent of primary SCADA/EMS infrastructure. Additionally, the two converter stations must maintain a backup inter-terminal channel architecturally independent of the primary inter-terminal control telecommunications link. Acceptable channels: dedicated licensed radio, satellite communications, or physically diverse fiber. Cellular backup does not satisfy CB-1 under MA conditions.
- **CB-2 (Local authority):** Converter station operators must have pre-authorized authority to execute all FC3 emergency response procedures — including converter block, monopolar operation initiation, and protective shutdown — without CntlCo direction and without inter-terminal coordination, under conditions where both the CntlCo and the remote converter station are unreachable.
- **CB-3 (Command Broker continuity):** MA communications degradation does not suspend the Command Broker obligation for above-Bright-Line Gen AI advisory tools. The qualified HVDC system operator or designated CntlCo authority on duty remains the mandatory Command Broker. Above-Bright-Line Gen AI power order recommendations must not be actioned without licensed operator confirmation, regardless of communications status. The FC3-specific Command Broker constraint from §3.6 applies: the Command Broker must evaluate the AC frequency trajectory at both terminals before authorizing any Gen AI-recommended power order change during a grid emergency.
- **CB-4 (Verification cadence):** All backup communications channels — both CntlCo-facing and inter-terminal — must be tested at an interval specified in the facility's design basis implementation. The test condition must simulate the MA baseline: primary channels unavailable at both converter stations simultaneously.
- **CB-5 (Degraded-mode procedures):** FC3 emergency operating procedures must include a degraded-communications mode executable at each converter station independently when both CntlCo contact and inter-terminal contact are simultaneously unavailable. Procedures must address: (a) autonomous converter station actions when all communications are lost; (b) backup channel activation sequence; (c) safe operating state maintained pending communications restoration; (d) threshold for controlled converter block rather than continued degraded-communications operation.

Section 6 — Consequence Analysis

6.1 Consequence Chain: SE-2 / TCI-Maximum Governing Case

Initiating event. Precision physical attack on the converter transformer at one or both converter stations damages the external oil system, initiating transformer shutdown. Protection systems de-energize the affected converter.

Tier 1 — Facility. Attacked converter transformer removed from service. HVDC transfer capability reduced to zero (bipolar attack) or approximately 50% (monopolar attack). TCI begins. Converter transformer replacement initiated; RT_HVLLC of 12–18 months governs restoration timeline.

Tier 2 — Grid. The receiving AC system experiences an instantaneous power deficit equal to the lost HVDC transfer block. Automatic generation control and governor response begin frequency recovery. If the deficit exceeds the frequency response capability, UFLS relays operate to shed load. For the duration of TCI, the receiving interconnection operates without its HVDC import resource.

Tier 3 — Cross-sector. Load shedding removes power from critical facilities in the receiving region. Consequence escalates from temporary outage to sustained deficit to potential mission degradation if TCI extends beyond backup generation endurance.

Extended consequence regime. Receiving interconnection manages reliability obligations without HVDC import contribution for the TCI duration. Resource adequacy margins reduced. Cascading risk following subsequent contingency events was elevated for the full TCI period.

6.2 MA Extension Consequence Chain

The MA extension consequence chain differs from the non-adversarial chain in the following ways. Intentionality extends the consequence from a single-terminal event to a coordinated attack on both terminals simultaneously, eliminating monopolar fallback, and times the attack to peak import dependency. Persistence is provided by supply chain interdiction, extending TCI beyond RT_HVLLC without further physical presence at the facility. Precision targets the converter transformer specifically, rather than the DC line or thyristor valves, and also targets attack timing. Denial prevents damage assessment, delays replacement equipment delivery, and may extend to area denial, preventing contractor mobilization.

Section 7 — Acceptance Criteria

7.1 Tier 1 Acceptance Criteria

The facility must sustain monopolar operation following loss of one pole's converter transformer, maintaining at least 50% of design transfer capacity during the monopolar period. Station service continuity must be maintained for a minimum of 72 hours through UPS or dedicated auxiliary generation. Converter transformer physical hardening must reduce the probability of replacement-class external damage from the design basis physical attack to an acceptable level.

7.2 Tier 2 Acceptance Criteria

The receiving AC system must demonstrate through NERC TPL planning analysis that it can maintain reliability obligations following instantaneous loss of the HVDC link at maximum transfer rate under all pre-contingency operating conditions within the TPL planning envelope. The receiving interconnection must demonstrate resource adequacy without HVDC import contribution for a duration equal to the maximum credible TCI — RT_HVLLC (12–18 months) — for all seasonal demand periods. Resource adequacy assessment must treat HVDC import as a contingency resource, not a firm resource.

7.3 Tier 3 Acceptance Criteria

The maximum duration of power interruption to any defense-critical or other highest-priority critical facility in the receiving region must not exceed that facility's backup power posture duration before mission-unrecoverable consequences occur. Where the most time-constrained critical facility has a backup power posture shorter than the maximum credible TCI, the Tier 3 criterion must be satisfied through alternative supply arrangements — dedicated mobile generation contracts, pre-positioned fuel reserves, or priority restoration agreements — established before an FC3 event occurs and available under MA access denial conditions.

Section 8 — Deferred Items

Item	Description	Resolution Criterion
DI-1	Multi-terminal HVDC scope. This document addresses point-to-point HVDC as the primary case. Multi-terminal HVDC systems present additional complexity: a fault or attack at one terminal affects all other terminals simultaneously.	Multi-terminal HVDC scope extension added to FC3 instrument or addressed in a dedicated annex following completion of the point-to-point analysis.
DI-2	Converter transformer strategic reserve gap. Unlike large power transformers for AC substations, converter transformers are not covered by STEP. No national strategic reserve for HVDC converter transformers exists, and no institutional owner for this gap has been identified.	Policy analysis completed; federal agency ownership identified; cost-benefit framework for strategic reserve established and presented to DOE and FERC.
DI-3	TCI acceptance criterion calibration. The specific TCI duration threshold above which regional resource adequacy cannot be maintained without unrecoverable cross-sector consequences requires quantitative analysis against each FC3 facility's specific receiving-system load profile.	TCI acceptance criterion calibrated for each FC3 facility as part of site-specific design basis implementation.
DI-4	GMD postulated event quantification. SE-4 identifies geomagnetic disturbance as the governing natural hazard. Quantitative GIC analysis requires site-specific geomagnetic latitude data, soil conductivity models, and DC resistance parameters for the converter transformer windings.	Quantitative GIC analysis completed for each FC3 facility consistent with NERC GMD standards TPL-007.
DI-5	Dual-DB AI Governance interaction points. CLOSED v0.2. Bright Line positions and Command Broker requirements established in §3.6: HVDC control optimization ML (below Bright Line with operating envelope boundary requirement), anomaly detection ML (below Bright Line with adversarial input robustness requirement), Gen AI HVDC operator support (above Bright Line, Command Broker required with AC frequency evaluation constraint).	CLOSED. Resolved in §3.6, v0.2.
DI-6	MA applicability four-filter block and communications design basis. CLOSED v0.3. Four-filter structured assessment established in §5.1; MA communications design basis CB-1 through CB-5 instantiated in §5.5, including FC3-specific inter-terminal backup channel obligation.	CLOSED. Resolved in §5.1 and §5.5, v0.3. Addresses N-15 and N-16 batch pass requirements per DBA-ES-MA-SF v0.2 OI-8 and OI-1.

Section 9 — References

Regulatory and Standards References

Federal Energy Regulatory Commission. Order No. 693: Mandatory Reliability Standards for the Bulk-Power System. Washington, D.C.: FERC, 2007.

North American Electric Reliability Corporation. FAC-001-3: Facility Connection Requirements. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. FAC-002-3: Facility Connection Requirements. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. TPL-001-5: Transmission System Planning Performance Requirements. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. TPL-007-4: Transmission System Planned Performance for Geomagnetic Disturbance Events. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

Government and Policy References

Defense Science Board Task Force on DoD Energy Strategy. “More Fight — Less Fuel.” Washington, D.C.: Office of the Under Secretary of Defense, February 2008.

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U.S. Department of Energy and North American Electric Reliability Corporation. “Final Report on the August 14, 2003 Blackout in the United States and Canada.” Washington, D.C.: DOE, April 2004.

Technical References

CIGRE Working Group B4.62. “Connectionless DC Grids.” CIGRE Technical Brochure 533. Paris: CIGRE, 2013.

Manitoba Hydro. “Bipole III Reliability Project.” Project documentation, Manitoba Hydro, Winnipeg, 2021.

Mohan, Ned, Tore M. Undeland, and William P. Robbins. Power Electronics: Converters, Applications, and Design. 3rd ed. Hoboken: John Wiley & Sons, 2003.

Series Parent and Cross-Reference Documents

Roxey, Tim. DBA-EN-SF: Energy Sector Framework. Version 0.8 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Version 0.9 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-MA-SF: Electric Sector Military Action Sector Framework. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-MA-QB: Electric Sector Military Action Quantitative Design Basis. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-TX-FC2: Large EHV Transmission Substation. Version 0.3 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-HY-FC1: Large Hydroelectric Storage Facilities. Version 0.4 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-EN-UC-2: Coordinated Multi-Node Attack on an Interconnection. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. RT_HVLLC Strategic Reserve White Paper. Version 0.2 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. GenMix Optimization Whitepaper. Version 0.3 — DRAFT. Eclectic Technologies, 2026.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial working draft. First publication under DBA-ES sector parent (v0.6). Established facility class definition (≥2,000 MW point-to-point HVDC, cross-interconnection role). Introduced TCI (Transfer Capability Isolation) as the governing FC3 consequence-duration metric. Retained RT_HVLLC for converter transformer replacement as a secondary metric. Instantiated all four sector-level postulated event categories. Developed DBA non-adversarial consequence envelope. Applied DB→DBA→DBA-MA three-layer progression. Incorporated the Nelson River system as a consequence model and the 2003 Northeast Blackout as a cascade precedent.
0.2	June 2026	T. Roxey	Full three-layer instrument. Parent updated to DBA-ES v0.6. Dual-DB AI Governance interaction points established in §3.6: HVDC control optimization ML (below Bright Line with operating envelope boundary requirement), anomaly detection ML (below Bright Line with adversarial input robustness requirement), Gen AI HVDC operator support (above Bright Line, Command Broker required with AC frequency evaluation constraint). DI-5 closed.
0.3	June 2026	T. Roxey	N-15/N-16 batch pass. Three changes: (1) Parent updated from DBA-ES v0.6 to v0.9; series grandparent DBA-EN-SF v0.8 added to cover and §1.6. (2) §5.1 MA Applicability four-filter structured assessment added — all four filters satisfied at class level; FC3 identified as the highest Filter 4 priority in DBA-ES series (member of both transmission pattern attack combinations). Closes N-15/OI-8 for FC3. (3) §5.5 MA Communications Design Basis added — CB-1 through CB-5 instantiated for FC3 with FC3-specific inter-terminal backup channel obligation. Closes N-16/OI-1 for FC3. DI-6 added and marked CLOSED. DBA-EN-SF v0.8, DBA-ES-MA-SF v0.2, DBA-ES-MA-QB v0.1 added to §9 references; DBA-ES updated to v0.9; FC2 updated to v0.3; HY-FC1 updated to v0.4. Revision history moved to the end of the document.